Streaming

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Streaming allows users to enjoy music or video instantly, on demand, as long as they have a reliable network connection. It's a convenient alternative to downloading, and a good option for devices with limited memory.

How it works

Streaming works by breaking the streamed media into small pieces of data. These are sent over the network in a structured manner that allows the user's machine to reconstruct the media second by second. Once a segment of the media is played, its corresponding data is thrown away. This process is similar to dipping a hand in a stream – new water continuously flows past the hand, but once the water leaves, it's gone.

"The **growth** [of users] over the next **10 years** will be in **streaming**."
Reed Hastings (b. 1960), American co-founder of Netflix

Bandwidth
Bandwidth is the amount of data that can flow into a network. In other words, it's the number of bits a computer receives per second.

TCP vs UDP
TCP and UDP are protocols for transmitting data. Unlike UPD, TCP establishes a link between computers, making it slower but more reliable and secure.

Unicast vs Multicast
Multicast allows a signal to be sent to multiple devices at once. Unicast, on the other hand, limits each transmission to a single receiver.

Buffering
When streaming, a computer stores a bit of data ahead of what is already playing. Buffering prevents irregular, jerky playback by controlling the rate of live streams.